

ME101: Engineering Mechanics, 2023-24

IIT Guwahati

Tutorial – 11 (Solution)

Problem1:

A 10-kg wooden plate is suspended from a pin support at A and is initially at rest. One 5kg metal sphere is released from rest at B and falls into a hemispherical cup C attached to the plate at the same level as the mass center G. Assuming that the impact is perfectly plastic, determine (a) the angular velocity of the plate and (b) the velocity of the mass center G of the plate immediately after the impact. You can consider dimensions L_1 , L_2 and L_3 as 0.8m, 1.6m, and 0.5m respectively. [10 Marks]

Solutions:

We have different dimensions as $L_1 = 0.8$ m, $L_2 = 1.6$ m, and $L_3 = 0.5$ m

Mass of the plate, $m_p = 10$ kg; mass of the sphere, $m_s = 5$ kg.

Mass moment of inertia of the square plate about the centroidal out-of-plane axis,

$$\bar{I} = \frac{1}{6} m_p L_2^2 = \frac{1}{6} (10) (1.6)^2 = 4.2667 \text{ kg-m}^2$$

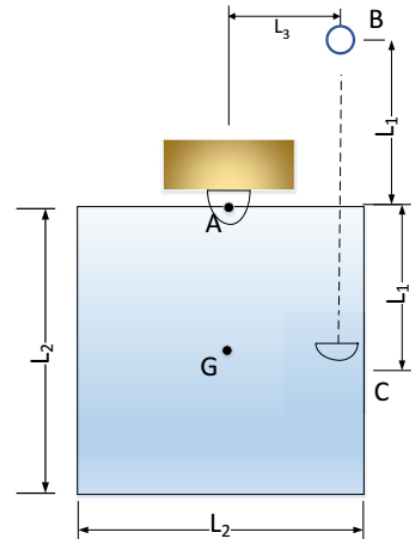


Figure 1: Problem1

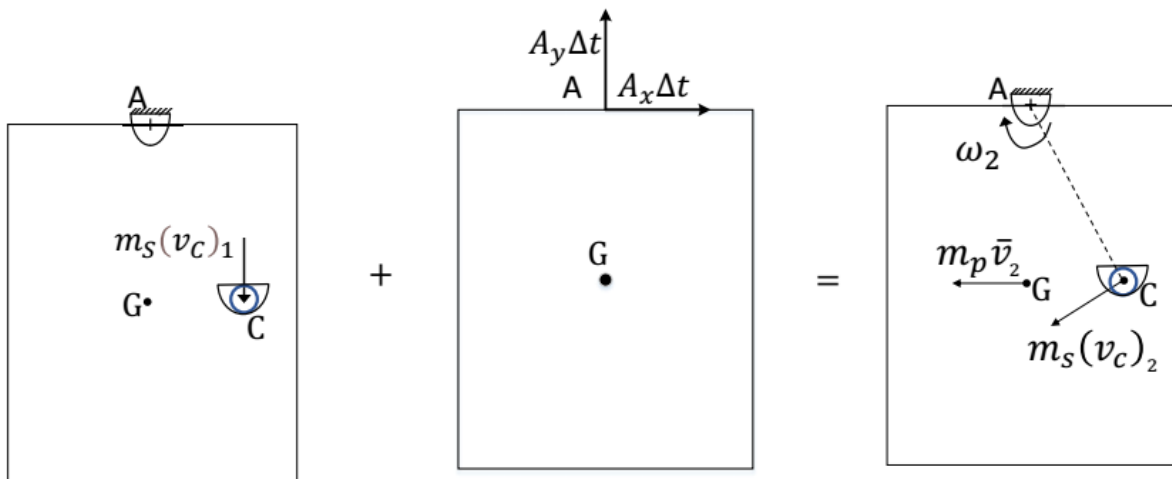


Figure 2

Velocity of sphere at location C just before the impact (state 1),

$$(v_c)_1 = \sqrt{2gy} = \sqrt{2(9.81)(1.6)} = 5.6028 \text{ m/s}$$

Impact analysis: as the impact is plastic, after the impact the metal sphere will not bounce, it will stick to C.

$$\text{Distance, AC} = \sqrt{0.5^2 + 0.8^2} = 0.9433 \text{ m}$$

Let, after the impact (state 2) the plate will have clockwise angular velocity as ω_2 .

Then, $(v_C)_2 = 0.9433 \omega_2$ (Direction will be perpendicular to AC as shown in Fig. 2).

Velocity of centroid of the plate, $\bar{v}_2 = 0.8\omega_2$

At state 1, initial angular momentum is contributed only by the sphere as the plate is static initially.

At state 1, total initial angular momentum with respect to point A (CW positive),

$$(H_A)_1 = (H_A)_{1\text{Plate}} + (H_A)_{1\text{Sphere}} = (H_A)_{1\text{Sphere}} = m_s(v_C)_1(0.5 \text{ m}) = 14.007 \text{ kgm}^2\text{s}$$

[3 Marks]

At state 2, final angular momentum of the plate with respect to the mass center,

$$(H_G)_{2\text{Plate}} = \bar{I}\omega_2 = 4.2667\omega_2$$

$$\text{Now, } (H_A)_{2\text{Plate}} = (H_G)_{2\text{Plate}} + 0.8m_p\bar{v}_2 = 4.2667\omega_2 + 6.4\omega_2 = 10.6667\omega_2$$

Also, for the sphere, angular momentum at state 2 (CW),

$$(H_A)_{2\text{Sphere}} = 0.9433 m_s(v_C)_2 = 4.4491\omega_2 \text{ kgm}^2\text{s}$$

At state 2, total angular momentum with respect to point A (CW positive),

$$(H_A)_2 = (H_A)_{2\text{Plate}} + (H_A)_{2\text{Sphere}} = 10.6667\omega_2 + 4.4491\omega_2 = 15.1158\omega_2 \text{ [3 Marks]}$$

The only external forces acting are reaction forces at A. So, Total angular impulse with respect to A from state 1 to state 2, $\sum M_{A1 \rightarrow 2} = 0$. [1 Marks]

Now, we can write the angular impulse momentum principle as,

$$(H_A)_1 + \sum M_{A1 \rightarrow 2} = (H_A)_2$$

$$\text{Or, } 14.007 + 0 = 15.1158\omega_2$$

$$\text{Or, } \omega_2 = 0.9266 \text{ rad/s}$$

Therefore, angular velocity of the plate is 0.9266 rad/s (CW) [2 Marks]

velocity of the mass center

$$\bar{v}_2 = 0.8\omega_2 = 0.7413 \text{ m/s (towards left). [1 Marks]}$$

Problem 2:

Figure 3 shows the Saturn V mobile launch platform A together with the umbilical tower B, unfueled rocket C, and crawler-transporter D which carries the system to the launch site. The approximate dimensions of the structure and locations of the mass centers G are given. The approximate masses are $m_A = 5 \text{ Gg}$, $m_B = 7 \text{ Gg}$, $m_C = 0.5 \text{ Gg}$, and $m_D = 3 \text{ Gg}$. The minimum stopping distance from the top speed of 3 km/h is 0.15 m . Compute the vertical component of the reaction under the front crawler unit F during the period of maximum deceleration. Considering $h_2 = 80 \text{ m}$, $h_3 = 50 \text{ m}$, $L_1 = L_2 = 15 \text{ m}$. [10 Marks]

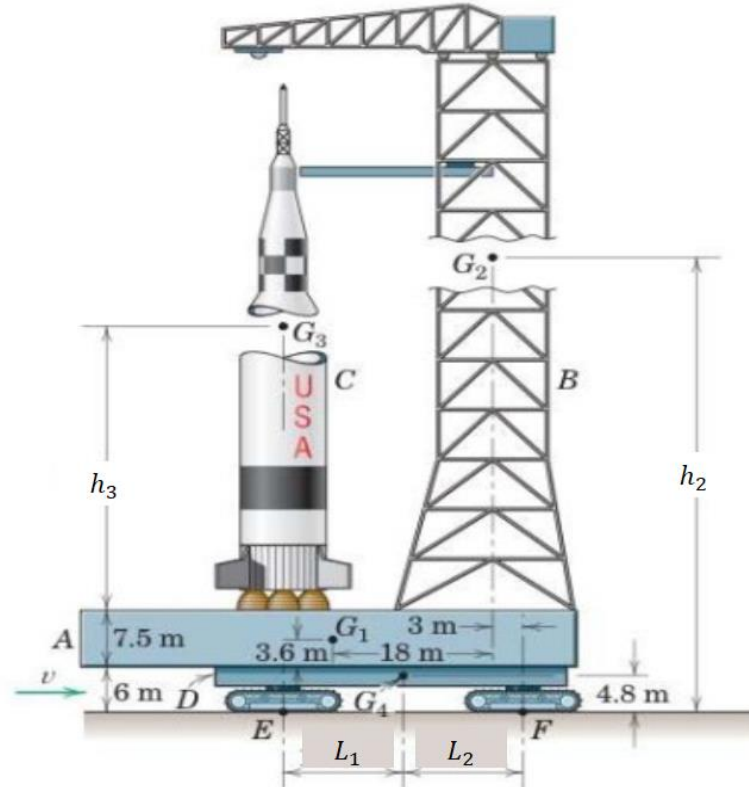


Figure 3: Problem2

Solution:

Given, $h_2 = 80 \text{ m}$, $h_3 = 50 \text{ m}$, $L_1 = L_2 = 15 \text{ m}$,

$m_A = 5 \text{ Gg}$, $m_B = 7 \text{ Gg}$, $m_C = 0.5 \text{ Gg}$, and $m_D = 3 \text{ Gg}$.

Let resultant mass center is at O and E is the origin of coordinate

$$\therefore x_g = \frac{(9 \times 5) + (15 \times 3) + (27 \times 7) + (0.5 \times 0)}{(5 + 3 + 7 + 0.5)} \text{ m} = 18 \text{ m},$$

$$\text{and } y_g = \frac{(5 \times 9.6) + (7 \times 80) + (0.5 \times 63.5) + (3 \times 4.8)}{15.5} \text{ m} = 42.2 \text{ m}$$

$$\text{Deceleration: } a = \frac{\left(\frac{3 \times 1000}{3600}\right)^2}{2 \times 0.15} \text{ m/s}^2 = 2.315 \text{ m/s}^2 \quad [5 \text{ Marks}]$$

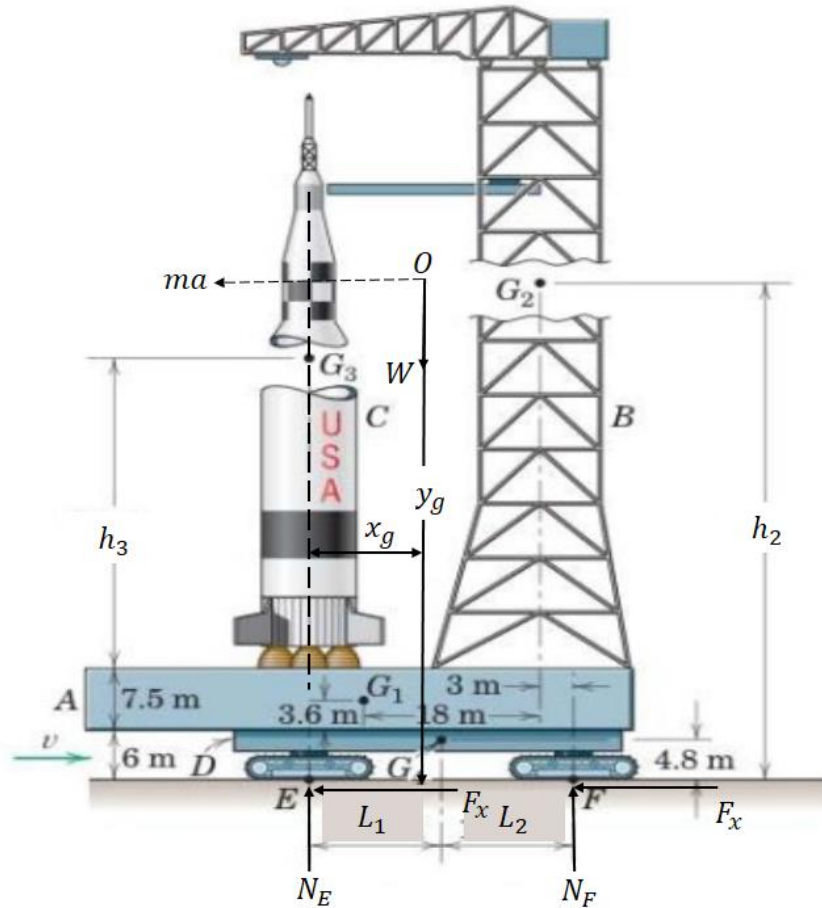


Figure 4

$$\sum M_E = ma \cdot d \quad (\curvearrowright +)$$

$$\Rightarrow (N_F \times 30) - (W \times x_g) = may_g$$

$$\Rightarrow N_F = \frac{(W \times x_g) + (may_g)}{30} \text{ N}$$

$$\Rightarrow N_F = \frac{(15.5 \times 9.81 \times 18) + (15.5 \times 2.315 \times 42.2)}{30} \text{ MN} = 141.71 \text{ MN} \quad [5 \text{ Marks}]$$

∴ vertical component of reaction under the front crawler unit F during the period of maximum deceleration will be 141.71 MN.

Problem 3:

The block A and attached rod have a combined mass of 10 kg and are confined to move along the 60° guide under the action of the 120 N applied force. The uniform horizontal rod has a mass of 4 Kg and length 0.6 m is welded to the block at B. Friction in the guide is negligible. Compute the bending moment M exerted by the weld on the rod at B. [10 Marks]

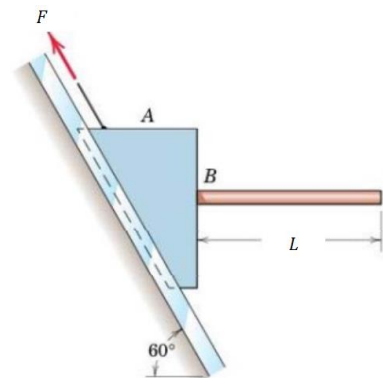


Figure 5: Problem 3

Solution:

Given, $L = 0.6 \text{ m}$, $F = 120 \text{ N}$.

And Combined mass of rod and block (m) = 10 Kg, mass of rod = 4 kg

$$\therefore \sum F_y = ma$$

$$\Rightarrow 120 - (10 \times 9.81 \times \sin 60^\circ) = 10a$$

$$\Rightarrow a = 3.5 \text{ m/s}^2 \quad [5 \text{ Marks}]$$

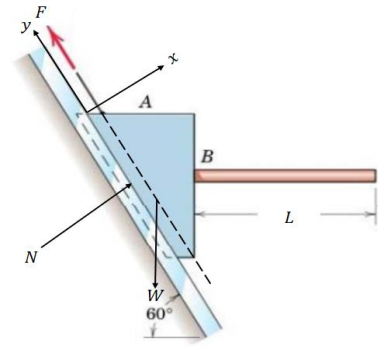


Figure 6

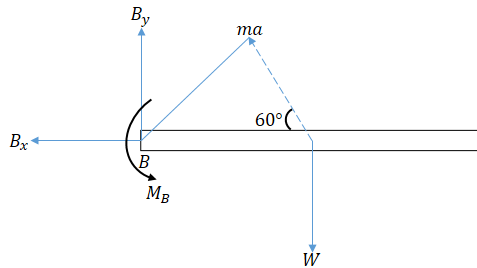


Figure 7

$$\sum M_B = ma \cdot d$$

Here, we take counter clockwise moment as a positive moment.

$$\Rightarrow M_B - (W \times 0.3) = 4 \times 3.5 \times 0.3 \sin 60^\circ$$

$$\Rightarrow M_B = [(4 \times 9.81 \times 0.3) + (4 \times 3.5 \times 0.3 \sin 60^\circ)] \text{ Nm}$$

$$= 15.41 \text{ Nm} \quad [5 \text{ Marks}]$$

$\therefore$ Bending moment at B = 15.41 Nm [Counter clockwise] (Ans)

Problem 4:

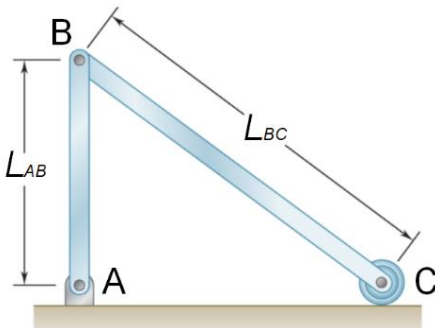


Figure 8: Problem 4

The uniform rods AB and BC of masses 10 kg and 15 kg, respectively, and the small wheel at C is of negligible mass. Knowing that in the position shown the velocity of wheel C is 10 m/s to the right, determine the velocity of pin B after rod AB has rotated through 90° . Here $L_{AB} = 1.2 \text{ m}$ and $L_{BC} = 2 \text{ m}$. [10 Marks]

Solution:

Moments of inertia:

$$\text{Rod AB: } I_A = \frac{1}{3} m_{AB} L_{AB}^2 = \frac{1}{3} (10)(1.2)^2 = 4.8 \text{ Kg. m}^2$$

$$\text{Rod BC: } \bar{I} = \frac{1}{12} m_{BC} L_{BC}^2 = \frac{1}{12} (15)(2)^2 = 5 \text{ Kg. m}^2$$

Position 1. As shown with bar AB vertical. Point G is the midpoint of BC.

$$V_1 = m_{AB} g h_{AB} + m_{BC} g h_{BC} = (10)(9.81)(0.6) + (15)(9.81)(0.6) = 147.15 \text{ J} \quad [2 \text{ Marks}]$$

Kinematics: At the instant shown in Position 1, $\omega_{BC} = 0$

$$\bar{v} = v_G = v_B = v_C = 10 \text{ m/s,}$$

$$\omega_{AB} = \frac{v_B}{L_{AB}} = \frac{10}{1.2} = 8.34 \text{ rad/s}$$

$$\begin{aligned} T_1 &= \frac{1}{2} I_A \omega_{AB}^2 + \frac{1}{2} m_{BC} \bar{v}^2 + \frac{1}{2} \bar{I} \omega_{BC}^2 \\ &= \frac{1}{2} (4.8)(8.34)^2 + \frac{1}{2} (15)(10)^2 + 0 \\ &= 916.93 \text{ J} \end{aligned} \quad [3 \text{ Marks}]$$

Position 2. Rod AB is horizontal.

$$V_2 = 0$$

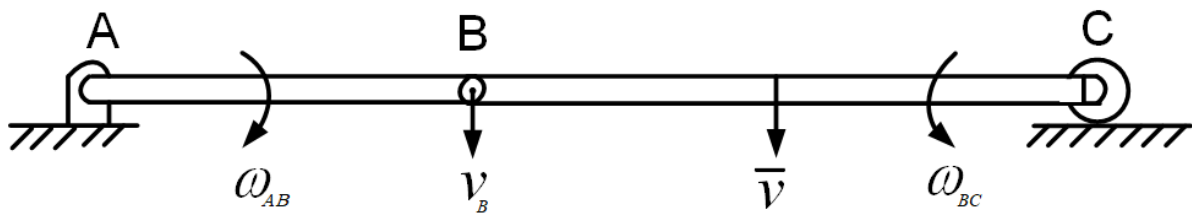


Figure 9

$$\begin{aligned} \omega_{AB} &= \frac{v_B}{L_{AB}} = \frac{v_B}{1.2} \\ \omega_{BC} &= \frac{v_B}{L_{BC}} = \frac{v_B}{2}, \end{aligned}$$

$$\bar{v} = \frac{1}{2} v_B$$

$$\begin{aligned} T_2 &= \frac{1}{2} I_A \omega_{AB}^2 + \frac{1}{2} m_{BC} \bar{v}^2 + \frac{1}{2} \bar{I} \omega_{BC}^2 \\ &= \frac{1}{2} (4.8) \left(\frac{v_B}{1.2} \right)^2 + \frac{1}{2} (15) \left(\frac{1}{2} v_B \right)^2 + \frac{1}{2} (5) \left(\frac{v_B}{2} \right)^2 \\ &= 4.167 v_B^2 \end{aligned} \quad [3 \text{ Marks}]$$

Conservation of energy. $T_1 + V_1 = T_2 + V_2$: $916.934 + 147.15 = 4.167 v_B^2$
 $\Rightarrow v_B = 15.98 \text{ m/s}, \Rightarrow v_B = 15.98 \text{ m/s} \downarrow$ [2 Marks]

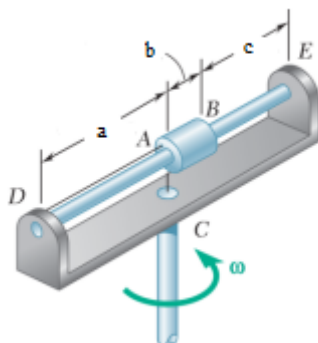


Figure 10: Problem 5

Problem 5:

A 1 kg tube AB can slide freely on rod DE, which in turn can rotate freely in a horizontal plane. Initially the assembly is rotating with an angular velocity $\omega = 10 \text{ rad/s}$ and the tube is held in position by a cord. The moment of inertia of the rod and bracket about the vertical axis of rotation is 0.50 kg.m^2 and the centroidal moment of inertia of the tube about a vertical axis is 0.003 kg.m^2 . If the cord suddenly breaks, determine (a) the angular velocity of the assembly after the tube has moved to

end E, (b) the energy lost during the plastic impact at E, (c) sliding velocity of tube AB just before the impact. Assume dimensions are $a= 300$ mm, $b= 100$ mm, $c=200$ mm. [10 Marks]

Solution:

Let Point C be the intersection of axle C and rod DE. Let Point G be the mass centre of tube AB.

Masses and moments of inertia about vertical axes.

$$m_{AB} = 1 \text{ kg}$$

$$\bar{I}_{AB} = 0.003 \text{ kg.m}^2$$

$$\bar{I}_{DCE} = 0.50 \text{ kg.m}^2$$

State 1: Initial state (before breaking of chord)

$$(r_{G/A})_1 = \frac{1}{2}(100) = 50 \text{ mm}$$

$$\omega_1 = 10 \text{ rad/s}$$

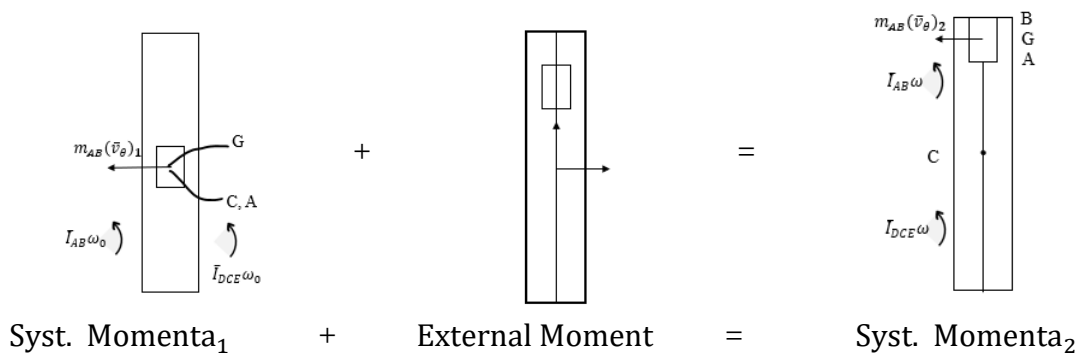
State 2: Just after plastic impact at E

$$(r_{G/A})_2 = 300 - 50 = 250 \text{ mm}$$

$$\omega = \omega_2$$

Kinematics:

$$(v_G)_\theta = \bar{v}_\theta = r_{G/C}\omega$$



Moment about C:

$$\bar{I}_{AB}\omega_1 + \bar{I}_{DCE}\omega_1 + m_{AB}(\bar{v}_\theta)_1(r_{G/C})_1 + 0 = \bar{I}_{AB}\omega_2 + \bar{I}_{DCE}\omega_2 + m_{AB}(\bar{v}_\theta)_2(r_{G/C})_2$$

$$[\bar{I}_{AB} + \bar{I}_{DCE} + m_{AB}(r_{G/G})_1^2]\omega_1 = [\bar{I}_{AB} + \bar{I}_{DCE} + m_{AB}(r_{G/C})_2^2]\omega_2$$

$$[0.003+0.50+(1)(0.05)^2](10)=[0.003+0.50+(1)(0.250)^2]\omega_2$$

$$(0.5055)(10)=0.5655\omega_2$$

$$\omega_2 = 8.939 \text{ rad/s}$$

(a) Angular velocity after the plastic impact- 8.94 rad/s [5 Marks]

Kinetic Energy $T = \frac{1}{2} \bar{I}_{AB} \omega^2 + \frac{1}{2} \bar{I}_{DCE} \omega^2 + \frac{1}{2} m_{AB} \bar{v}^2$

$$T_1 = \frac{1}{2} (0.003) (10)^2 + \frac{1}{2} (0.50) (10)^2 + \frac{1}{2} (1) (0.05)^2 (10)^2$$

$$= 25.275 \text{ J}$$

$$T_2 = \frac{1}{2} (0.003) (8.939)^2 + \frac{1}{2} (0.50) (8.939)^2 + \frac{1}{2} (1) (0.25)^2 (8.939)^2$$

$$= 22.59334 \text{ J}$$

(b) Energy lost $T_1 - T_2 = 2.681 \text{ J}$ [3 Marks]

There is some sliding velocity of tube AB with respect to rod DE, let it is v_{sliding} .

After plastic impact $v_{\text{sliding}} = 0$, the tube AB remain in contact with end E.

Before the plastic impact there is no loss in kinetic energy.

So, $\text{Energy lost} = \frac{1}{2} m_{AB} v_{\text{sliding}}^2$

$$2.681 = \frac{1}{2} (1) (v_{\text{sliding}}^2)$$

$$v_{\text{sliding}} = 2.315 \text{ m/s}$$

(c) sliding velocity of tube AB just before the impact- 2.315 m/s [2 Marks]